

EUREKA, NU, Canada

WMO: 719170

Lat: 79.989N

Lon: 85.934W

Elev: 34

StdP: 14.68

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-54.1	-50.7	-54.8	0.2	-46.6	-52.1	0.3	-46.2	34.9	-10.5	27.5	-16.7	1.1	130	1.145

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	55.8	46.3	52.4	44.6	49.4	43.0	46.7	54.7	45.0	51.7	43.3	48.9	10.8	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
41.0	37.9	47.7	39.2	35.2	47.3	37.6	33.1	46.1	18.5	55.1	17.6	51.9	16.8	49.2	54.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
26.3	22.3	19.2	DB	-55.6	61.2	3.1	5.2	-57.8	64.9	-59.6	68.0	-61.4	70.9	-63.6	74.7
			WB	-51.8	49.7	2.5	2.7	-53.6	51.6	-55.0	53.2	-56.4	54.7	-58.2	56.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.2	-31.9	-34.1	-32.5	-14.5	14.6	37.6	43.9	38.4	22.1	-3.0	-17.9	-27.0		
	DBStd	30.72	10.81	11.04	12.34	11.86	10.62	6.11	5.00	5.73	8.90	13.20	12.20	12.60		
	HDD50	18321	2538	2356	2558	1935	1098	372	199	361	836	1644	2037	2387		
	HDD65	23783	3003	2776	3023	2385	1563	821	653	825	1286	2109	2487	2852		
	CDD50	13	0	0	0	0	0	1	11	1	0	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	6.7	5.0	4.6	4.2	4.9	7.5	10.7	10.8	9.5	7.8	5.0	4.8	4.9		
	PrecAvg	3.0	0.1	0.1	0.1	0.2	0.1	0.3	0.5	0.6	0.4	0.3	0.2	0.1		
	PrecMax	5.6	0.4	0.6	0.5	0.8	0.6	1.2	1.8	1.5	1.1	1.0	0.7	0.9		
(15) Precipitation	PrecMin	1.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	0.8	0.1	0.1	0.1	0.2	0.1	0.2	0.4	0.4	0.3	0.2	0.2	0.2		
	DB	3.3	2.5	8.8	21.0	37.8	57.2	63.5	55.1	38.7	29.0	18.5	15.9			
	MCWB	1.9	1.4	8.0	19.3	33.5	46.5	49.7	46.1	34.7	27.6	16.9	14.9			
	2% DB	-4.3	-5.7	0.3	14.3	34.7	51.8	58.4	51.2	36.2	24.9	12.8	6.0			
(21) Mean Coincident Wet Bulb Temperatures	MCWB	-5.3	-6.4	-0.6	12.9	31.5	43.4	47.5	44.5	33.4	23.7	11.5	4.9			
	5% DB	-10.2	-11.6	-8.0	7.0	31.3	48.3	54.8	48.2	34.4	21.0	6.8	-1.2			
	MCWB	-10.9	-12.4	-8.7	5.7	28.7	41.6	46.0	43.0	32.1	20.0	5.8	-2.1			
	10% DB	-15.7	-17.4	-13.9	2.0	28.3	45.5	51.9	45.8	32.7	16.4	0.6	-7.1			
	MCWB	-16.3	-17.9	-14.4	0.8	26.0	39.8	44.5	41.4	30.5	15.5	-0.4	-7.8			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	2.3	1.8	8.2	19.4	33.9	46.7	50.3	46.8	35.6	27.7	16.6	14.8			
	MCDB	3.3	3.0	9.1	21.1	37.1	56.4	61.8	53.1	37.9	28.8	18.1	15.9			
	2% WB	-4.9	-6.1	-0.5	13.0	31.6	43.7	47.9	45.1	33.9	23.8	11.6	4.9			
	MCDB	-4.0	-5.4	0.3	14.4	34.6	51.1	57.1	50.3	35.8	24.9	12.7	6.0			
	5% WB	-10.6	-11.7	-8.2	5.7	28.9	41.7	46.2	43.4	32.4	20.0	5.7	-2.0			
(31) Mean Daily Temperature Range	MCDB	-9.9	-11.1	-7.6	6.9	31.2	48.0	54.1	47.7	34.3	20.9	6.8	-1.2			
	10% WB	-15.9	-17.2	-14.0	0.9	26.1	39.9	44.7	41.7	30.6	15.5	-0.4	-7.8			
	MCDB	-15.3	-16.7	-13.4	2.0	28.3	45.1	51.5	45.6	32.5	16.5	0.6	-7.2			
	5% DB	11.0	11.3	10.6	12.0	9.6	7.7	9.7	6.8	7.2	11.0	11.2	10.8			
	MCDBR	17.8	18.4	17.1	14.0	9.0	12.8	14.9	10.5	6.2	10.7	15.9	16.6			
(36) Clear-Sky Solar Irradiance	MCWBR	17.4	17.9	16.9	13.4	7.3	7.7	8.5	6.3	4.8	10.2	15.3	16.0			
	5% WB	17.8	18.2	17.5	13.9	9.0	12.4	14.2	10.1	5.8	10.7	16.1	16.2			
	MCWBR	17.4	17.8	17.1	13.3	7.4	7.7	8.5	6.6	4.7	10.2	15.5	15.8			
	taub	N/A	N/A	0.208	0.254	0.285	0.318	0.320	0.296	0.216	N/A	N/A	N/A			
	taud	N/A	N/A	2.053	2.206	2.231	2.359	2.465	2.466	2.266	N/A	N/A	N/A			
(42) All-Sky Solar Radiation	Ebn at Noon	0	0	179	249	263	259	253	237	189	0	0	0			
	Edh at Noon	0	0	13	23	29	28	24	19	11	0	0	0			
(44) All-Sky Solar Radiation	RadAvg	0	0	221	1131	2083	2133	1781	1039	327	0	0	0			
	RadStd	0	0	27	66	99	179	165	103	25	0	0	0			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page